

Dhawal Modi

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Experience

Kaiser Permanente

Jul 2025 – Present

Machine Learning Engineer

Division of Research

- Trained a DeepLabV3+ model for E/A ratio measurement using a custom curated training dataset, reducing mean absolute error from 0.3414 to 0.2157 (~37% improvement) and decreasing common failure cases (9% to 3%).
- Built a scalable inference pipeline for a CAC segmentation model for ~1.5M DICOM CT scans, integrating DICOM discovery, TotalSegmentator based heart masking and post processing to reduce false-positive predictions.
- Improved CAC Segmentation model inference throughput with Python Multiprocessing and multi-gpu scheduling, reducing per-series inference time from 260s to 15s (~17x faster).
- Processed ~500K echocardiogram videos using Python and OpenCV and built a frame matching workflow to link video frame to clinician-annotated measurement images, enabling reliable use of labeled measurements in model development and QA.

University of California, Merced

Jan 2024 – Dec 2024

Graduate Student Researcher

MoCA Lab

- Finetuned ENet and SegFormer Semantic Segmentation models for Robot path traversability application on custom dataset with an mIOU of 0.83 and 0.729 respectively (improvement of 62% and 41% over baseline models).
- Engineered real-time optimization solutions for robotic systems, achieving 25ms inference time through efficient model deployment and hardware-software integration on Nvidia Orin AGX platform.

Tata Consultancy Services Limited

Nov 2020 – Jul 2023

Software Engineer

- Implemented and streamlined validation routines for Outward Direct Debits and Credit Transfer payment channels, developing interfaces as per FPS and HKICL specifications leading to high-efficiency processing, handling up to 100,000 transactions per minute.
- Designed Spring REST API Client integration with BPAY API, enhancing real-time bill payment processing by 35%.
- Optimized Oracle SQL database queries, reducing Direct Debit and Credit Transfer Payment file upload times by 15% and boosting overall application performance.

Technical Skills

Languages: Python, C++, CUDA, Java, TypeScript

Frameworks: PyTorch, OpenCV, Scikit, Huggingface, Pandas, Numpy, Django

Databases: Oracle SQL, PostgreSQL, IBM Db2

Technologies: ROS, XML, Shell Scripting, $\LaTeX$

Education

University of California - Merced

Master of Science in Electrical Engineering & Computer Science (GPA: 3.90 / 4.00)

Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Bachelor of Engineering in Electronics & Communications Engineering (GPA: 3.27 / 4.00)

Projects

LLM Prompt Recovery | Python, PyTorch, Accelerate, Transformers

- Developed and Finetuned Google Gemma 7b and Mistral Mixtral 7b LLMs using QLoRA and Huggingface PEFT to predict a "Rewrite Prompt" from the "Original Prompt" and the "Transformed Text" pairs.
- Achieved top 45% ranking (986/2186) in competition; best submission using Phi-2 model scored 0.6188, outperforming fine-tuned Mixtral 8x7B (0.6056) and Gemma 7B (0.5981) models.

Heart Activation time & Transmembrane Potential Reconstruction | Python, PyTorch, Scikit, Seaborn, Matplotlib

- Implemented Customized 1D SqueezeNet CNN model to predict Myocardium Activation Times and Transmembrane Potential Curve reconstruction with 97.46% accuracy on 1600 test samples